Scan Line

Instructor: Dinesh Maharjan

2022

Graphics Pipeline

- In computer graphics we render 2D images or scenes.
- This involves a set of stage which constitute the graphics pipeline.
- It includes 5 stages.
- Object representation
- Modeling Transformation
- Lighting or coloring
- Viewing pipeline
- Scan conversion or rendering

Object representation

- It defines the object that will be part of scene.
- It involves specifying the vertices or edges with respect to some reference frame.
- Objects are defined in its own or local coordinate.
- Objects can be cubes, spheres, cylinders etc.
- Objects shape, size and position is not important at the point of its representation.

LINE DRAWING

Description: Given the specification for a straight line, find the collection of addressable pixels which most closely approximates this line.

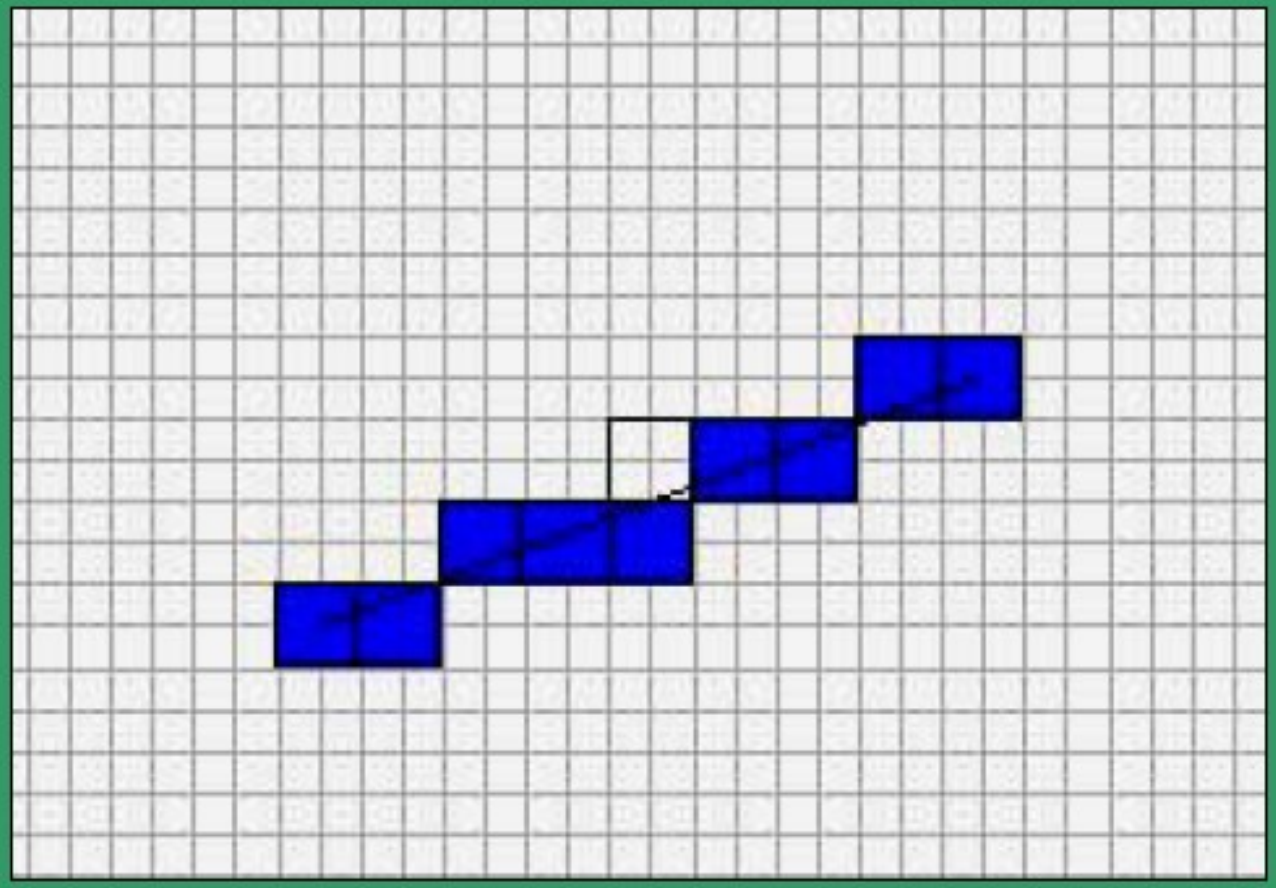
Goals (not all of them are achievable with the discrete space of a raster device):

- Straight lines should appear straight.
- Lines should start and end accurately, matching endpoints with connecting lines.
- Lines should have constant brightness.
- Lines should be drawn as rapidly as possible.

Problems:

- How do we determine which pixels to illuminate to satisfy the above goals?
- Vertical, horizontal, and lines with slope = +/- 1, are easy to draw.
- Others create problems: stair-casing/ jaggies/aliasing.
- Quality of the line drawn depends on the location of the pixels and their brightness

It is difficult to determine whether a pixel belongs to an object



Direct Solution:

Solve $y=mx+b$, where $(0,b)$ is the y-intercept and m is the slope.

Go from x_0 to x_1:

calculate round(y) from the equation.

Take an example, $b = 1$ (starting point $(0,1)$) and $m = 3/5$.

Then $x = 1, y = 2 = round(8/5)$

$x = 2, y = 2 = round(11/5)$

$x = 3, y = 3 = round(14/5)$

$x = 4, y = 3 = round(17/5)$

$x = 5, y = 4 = round(20/5)$

For results, see next slide.

About Slope

- Screen has only positive x axis and positive y axis.
- If abs(dx)>abs(dy), increment x by 1 and workout for y.
- Otherwise increment y by 1 and workout for x

Direct Method

- (x1,y1)=(2,2) and (x2,y2)=(7,5)
- dx=x2-x1=7-2=5
- dy=y2-y1=5-3=2
- m=dy/dx=2/5=0.4
- c=y-mx
- =2-0.4*2
- =1.2
- x=x1=2
- y=y1=2

i	x	y=mx+c	Round(y)	output
0	2	2	2	(2,2)
1	3	2.6	3	(3,3)
2	4	3.2	3	(4,3)
3	5	3.8	4	(5,4)
4	6	4.4	4	(6,4)
5	7	5	5	(7,5)

DDA

- Stands for Digital Differential Analyzer.
- An increment approach to speed up line scan conversion.
- It increments either x or y by 1.
- Then it increment or decrement another co-ordinate by slope or reciprocal of slope.
- The increment or decrement is determined by the direction of change in x or y.

DDA

- j

$$\text{if } |m| <= 1, x_{k+1} = x_k + 1 \text{ and } y_{k+1} = y_k + m$$
$$\text{else, } y_{k+1} = y_k + 1 \text{ and } x_{k+1} = x_k + 1/m$$

DDA Algorithm

- Let initial point (x1,y1) and final (x2, y2)
- dx= x2-x1
- dy=y2-y1
- if abs(dx)>abs(dy), step=abs(dx)
- Else step=abs(dy)
- xinc=dx/step
- yinc=dy/step

DDA Algorithm

- i=0
- While i<=step
- Drawpixel(round(x1),round(x2))
- X1=x1+xinc
- Y1=y1+yinc

DDA

- (x1,y1)=(2,2) and (x2,y2)=(7,5)
- dx=x2-x1=7-2=5
- dy=y2-y1=5-2=3
- Since abs(dx)>abs(dy),
- step=abs(dx)=5
- xinc=dx/step=5/5=1
- yinc=dy/step=3/5=0.6
- x=x1=2
- y=y1=2

i	x	y	output
0	2	2	(2,2)
1	3	2.6	(3,3)
2	4	3.2	(4,3)
3	5	3.8	(5,4)
4	6	4.4	(6,4)
5	7	5	(7,5)

DDA

- (x1,y1)=(7,5) and (x2,y2)=(2,2)
- dx=x2-x1=2-7=-5
- dy=y2-y1=2-5=-3
- Since abs(dx)>abs(dy),
- step=abs(dx)=5
- xinc=dx/step=-5/5=-1
- yinc=dy/step=-3/5=-0.6
- x=x1=7
- y=y1=5

i	x	y	output
0	7	5	(7,5)
1	6	4.4	(6,4)
2	5	3.8	(5,4)
3	4	3.2	(4,2)
4	3	2.6	(3,3)
5	2	2.0	(2,2)

DDA

- (x1,y1)=(2,3) and (x2,y2)=(1,9)
- dx=x2-x1=1-2=-1
- dy=y2-y1=9-3=6
- Since abs(dy)>abs(dx),
- step=abs(dy)=6
- xinc=dx/step=-1/6=-0.167
- yinc=dy/step=6/6=1
- x=x1=2
- y=y1=3

i	x	y	output
0	2.00	3	2,3
1	1.83	4	2,4
2	1.67	5	2,5
3	1.50	6	2,6
4	1.33	7	1,7
5	1.17	8	1,8
6	1.00	9	1.9

DDA

- Eliminates floating point multiplication m.x
- Any point can be initial point.
- Still has to perform float operations and round operations
- Line is not smooth

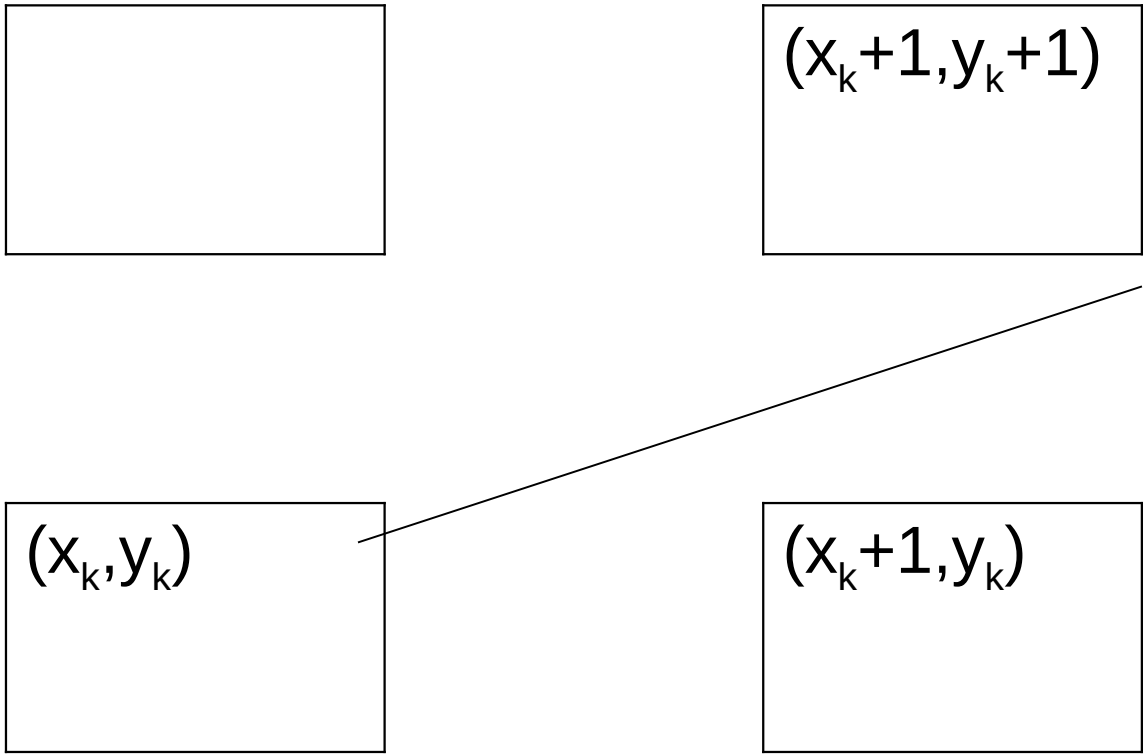
Bresenham

- It starts from the left end pixel
- It samples closest pixel to the line for each increment in x coordinate.
- If (x_1,y_1) is starting point and (x_2,y_2)
- It uses only integers to scan a line.

Bresenham

- Let current point be (x_k,y_k) and final point be (x_n,y_n)
- Let the equation of line be $y=mx+b$
- Since bresenham always increase x by 1,
- Next co-ordinate should be (x_k+1,y) where y is can be either y_k or y_k+1
- Now $y=m(x_k+1)+b$

Bresenham



Bresenham

- Distance between (x_k+1,y_k+1) and (x_k+1,y) is
- $d2=y_k+1-y$ (upper distance)
- Or $d2=y_k+1 - m(x_k+1) - b$
- Distance between (x_k+1,y_k) and (x_k+1,y) is
- $d1=y-y_k$ (lower distance)
- Or, $d1=m(x_k+1)+b-y_k$
- Let us evaluate d1-d2
- $d1-d2= m(x_k+1)+b-y_k - y_k -1+m(x_k+1)+b$
- $=2m(x_k+1)+2b -2y_k - 1$
- $dx(d1-d2)=2dy(x_k+1)+2dx.b – 2dx.y_k - dx$

Bresenham

- $dx(d1-d2)= 2dy(x_k+1)+2dx.b – 2dx.y_k - dx$
- $=2dy.x_k+2dy+2dx.b – 2dx.y_k - dx$
- Here, 2dy+2dx.b-dx is constant and does not contribute in decision making.
- So can be discarded
- So, $dx.(d1-d2)= 2dy.x_k – 2dx.y_k$
- Let $p_k=dx.(d1-d2) = 2dy.x_k – 2dx.y_k$
- Be current decision parameter and Next is
- $p_{k+1}=2dy.x_{k+1} –2 dx.y_{k+1}$

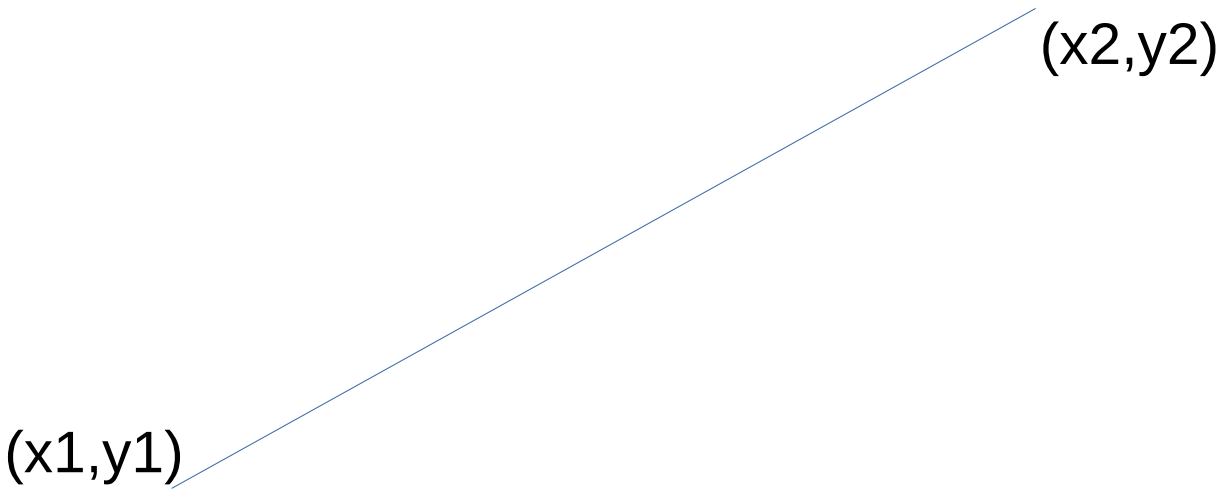
Bresenham

- If we see carefully dx cannot be negative since x2>x1.
- So, if $p_k<0$, then (x_k+1,y_k) is closer to the line, otherwise (x_k+1,y_k+1) is closer.
- If $p_k<0$, then $y_{k+1}=y_k$ and $x_{k+1}=x_k+1$
- So, $p_{k+1}=2dy.(x_k+1) - 2dx.y_k$
- Or $p_{k+1}= 2dy.x_k+ 2dy - 2dx.y_k$
- Or, $p_{k+1}= p_k +2dy$

Bresenham

- For initial decision parameter,
- $p_1 = 2dy - dx$
- This is midpoint theorem

Bresenham



- dx>dy
- Loop will be based on dx.
- x2>x1 and y2>y1
- So, x1 and y1 will increase to x2 and y2.
- Because, Bresenham always starts from (x1,y1)

Bresenham Pseudocode

- Let intial point be (x1,y1) and final point be (x2,y2)
- dx=abs(x2-x1)
- dy=abs(y2-y1)
- p=2.dy-dx
- x=x1
- y=y1
- i=0
- while(i<=dx)
- Drawpoint(x,y)

Bresenham Pseudocode

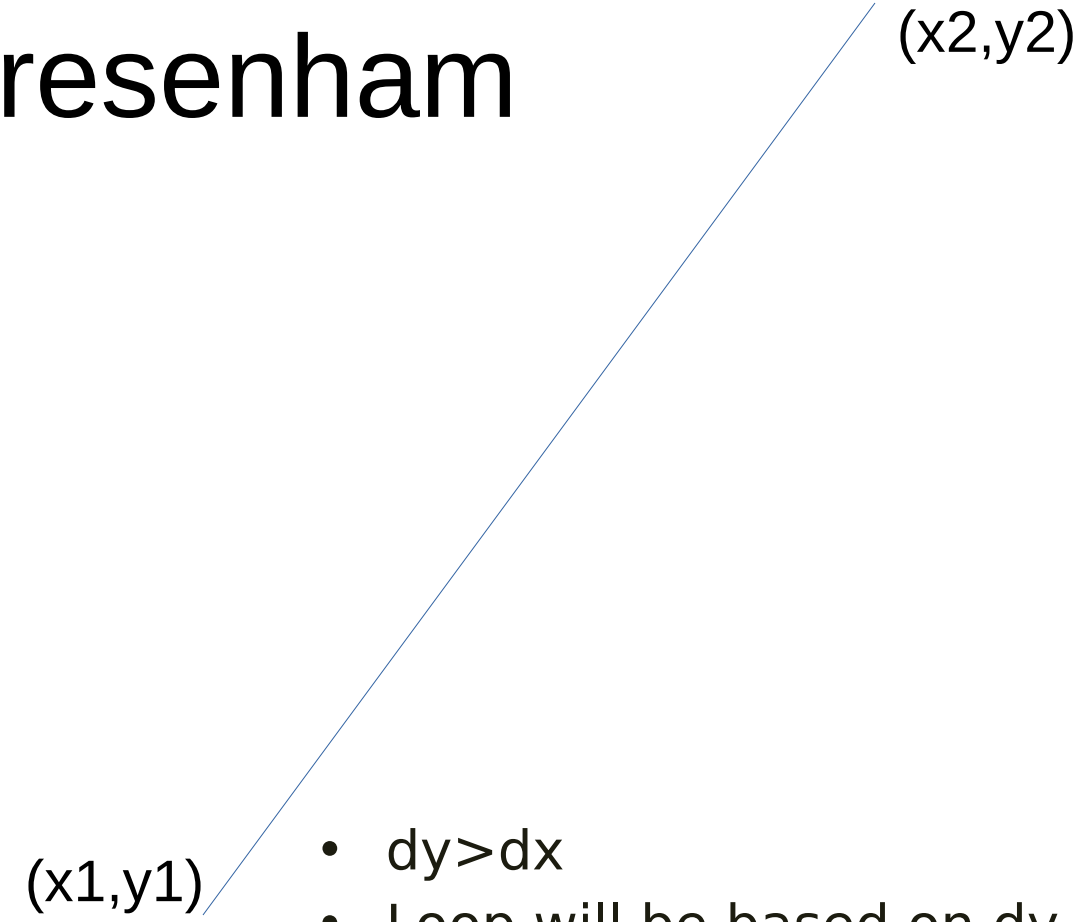
- X=x+1
- If p<0, p=p+2.dy
- Else y=y+1, p=p+2.dy-2dx

Bresenham Example

- (x1,y1)=(2,2) and (x2,y2)=(7,5)
- dx=abs(x2-x1)=abs(7-2)=5
- dy=abs(y2-y1)=abs(5-2)=3
- p=2.dy-dx=1
- x=x1=2
- y=y1=2

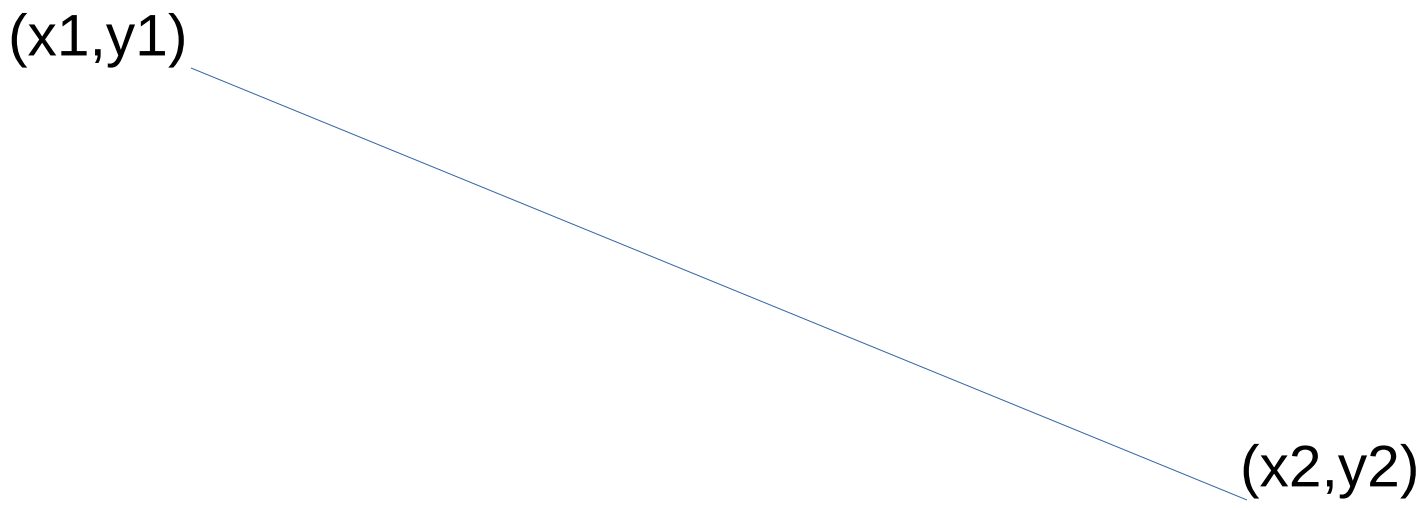
i	x	y	p	output
0	2	2	1	(2,2)
1	3	3	-3	(3,3)
2	4	3	3	(4,3)
3	5	4	-1	(5,4)
4	6	4	5	(6,4)
5	7	5	1	(7,5)

Bresenham



- dy>dx
- Loop will be based on dy.
- x2>x1 and y2>y1
- So, x1 and y1 will increase to x2 and y2.
- Because, Bresenham always starts from (x1,y1)

Bresenham



- dx>dy
- Loop will be based on dx.
- x2>x1 and y2<y1
- So, x1 will increase and y1 will decrease
- Because, bresenham always starts from (x1,y1) to (x2,y2)

Final Bresenham Pseudocode

- Let intial point be (x1,y1) and final point be (x2,y2)
- xinc=1 and yinc=1
- If x1>x2, then xinc=-1
- If y1>y2, then yinc=-1
- dx=abs(x2-x1)
- dy=abs(y2-y1)
- i=0

Final Bresenham Pseudocode

- if(dx>dy)
  - p=2.dy-dx
  - while(i<=dx)
    - draw(x,y)
    - x=x+xinc
    - If p<0, p=p+2dy
    - else, y=y+yinc, p=p+2dy-2dx

Final Bresenham Pseudocode

- if(dy>dx)
  - p=2.dx-dy
  - while(i<=dy)
    - draw(x,y)
    - y=y+yinc
    - If p<0, p=p+2dx
    - else, x=x+xinc, p=p+2dx-2dy

Bresenham

- All the computations are integer
- All floating points are removed
- No round function.
- Huge improvement in terms of speed of computation which is desirable
- Because screen is to be refreshed at very high speed to avoid flickr.

Drawing Circle

- Let as consider a circle with radius r and center at (0,0)
- The equation is $x^2+y^2=r^2$
- $F(x,y)=x^2+y^2-r^2$
- We can draw circle by computing y for each incremented value of x.
- But, this is very expensive.
- It has to perform square root operations.
- Moreover, the gap between computed pixels is not uniform.

Mid Point Theorem

- Similar to bresham algorithm, mid point theorem samples closest y-coordinate for each incremented x.
- For this it uses mid point of the next possible points.
- Since circle is has 8 symmetric points, it only samples points on first octant and plots remaining octants of the circle by the method of symmetry.

Drawing Circle

CIRCLE DRAWING

Only considers circles centered at the origin with integer radii.

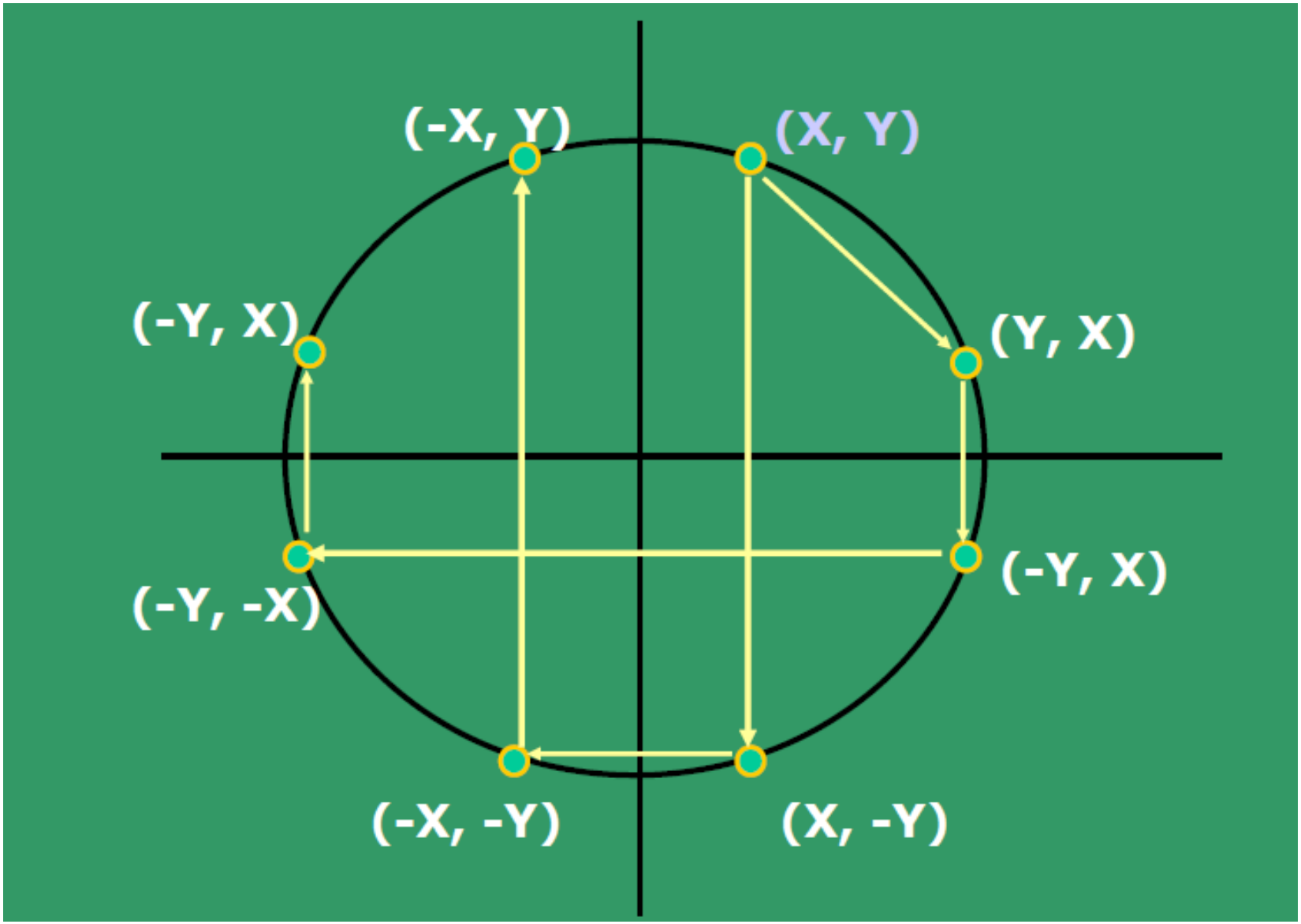
Can apply translations to get non-origin centered circles.

Explicit equation: $y = +/- \text{sqrt}(R^2 - x^2)$

Implicit equation: $F(x,y) = x^2 + y^2 - R^2 = 0$

Note: Implicit equations used extensively for advanced modeling

Drawing Circle



Drawing Circle

Use of Symmetry: Only need to calculate one octant. One can get points in the other 7 octants as follows:

```
Draw_circle(x, y)

begin
    Plotpoint (x, y);  Plotpoint (y, x);

    Plotpoint (x, -y); Plotpoint (-y, x);

    Plotpoint (-x, -y) ; Plotpoint (-y, -x);

    Plotpoint (-x, y); Plotpoint (-y, x);
end
```

MIDPOINT CIRCLE ALGORITHM

Will calculate points for the second octant.

Use *draw_circle* procedure to calculate the rest.

Now the choice is between pixels E and SE.

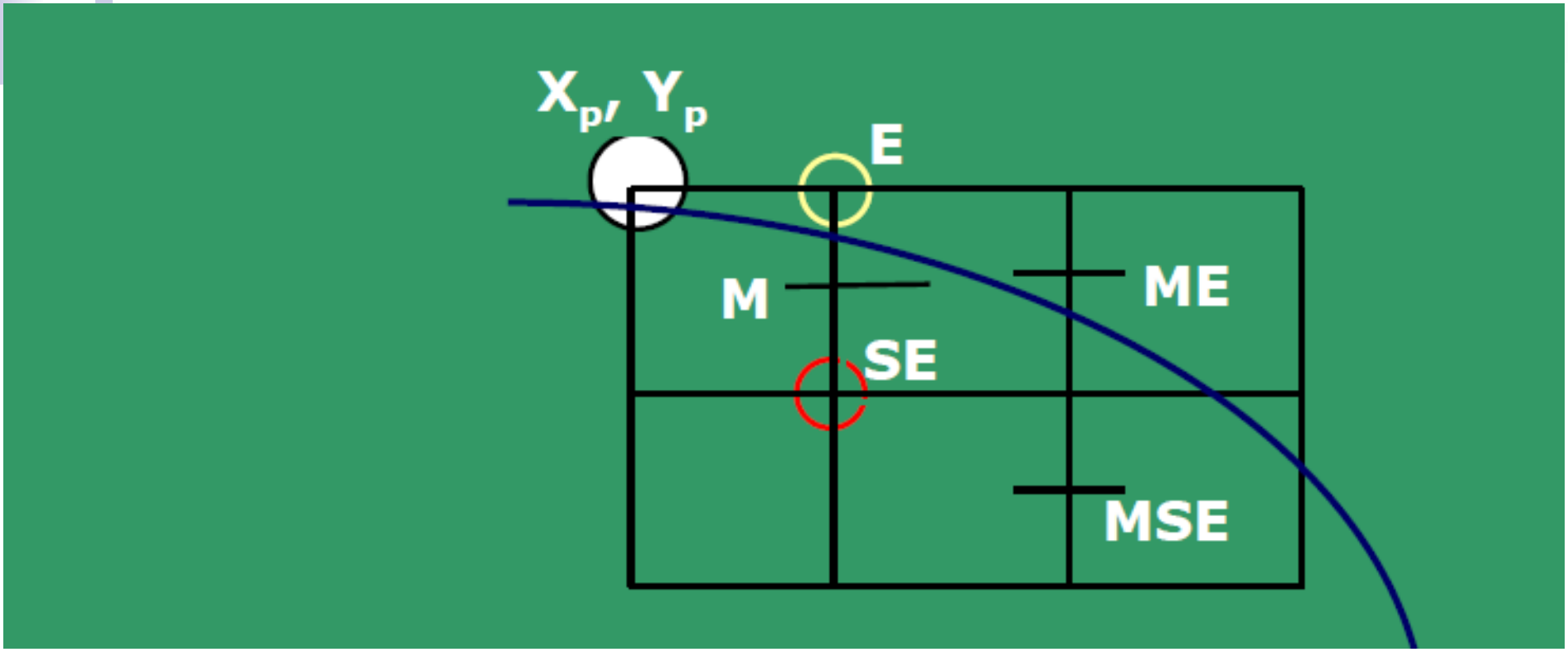
$F(x, y) = x^2 + y^2 - R^2 = 0$

$F(x, y) > 0$ if point is outside the circle

$F(x, y) < 0$ if point inside the circle.

Mid Point Theorem

- Let as consider a circle with radius r and center at (0,0)
- Let current point on circle be (x_k,y_k)
- Two possible next points are $E(x_k+1,y_k)$ and $SE(x_k+1,y_k-1)$
- Mid point of these two points are $M(x_k+1, y_k-0.5)$


X_p, Y_p
E
M
ME
SE
MSE

Mid Point Theorem

- If $F(x_p+1,y_p-0.5)>0$, M lies outside of circle.
- It means SE is closer to circle and we chose SE.
- If $F(x_p+1,y_p-0.5)<0$, M lies inside circle.
- It means E is closer to circle and we chose E.

Mid Point Theorem

- $F(x_p+1,y_p-1/2)=(x_p+1)^2+(y_p-1/2)^2 - r^2$
- Let decision parameter be
- $d_p= (x_p+1)^2+(y_p-1/2)^2 – r^2$
- $=x^2_p+2.x_p+1+y^2_p - y_p+1/4 - r^2$
- $d_{p+1}=x^2_{p+1}+2x_{p+1}+1+y^2_{p+1}-y_{p+1}+1/4 - r^2$

Mid Point Theorem

- If $d_p >= 0$, then
- $y_{p+1} = y_p - 1$ and $x_{p+1} = x_p + 1$
- $d_{p+1} = (x_p+1)^2 + 2(x_p+1) + 1 + (y_p-1)^2 - (y_p-1) + 1/4 - r^2$
- $= (x_p+1)^2 + 2(x_p+1) + 1 + y^2p - 2y_p + 1 - y_p + 1 + 1/4 - r^2$
- $= (x_p+1)^2 + y^2_p - y_p + 1/4 - r^2 + 2(x_p+1) + 1 - 2y_p + 1 + 1$
- $= (x_p+1)^2 + (y_p - 1/2)^2 - r^2 + 2(x_p+1) + 1 - 2y_p + 1 + 1$
- $= d_p + 2(x_p+1) + 1 - 2y_p + 1 + 1$
- $= d_p + 2x_p - 2y_p + 5$

Mid Point Theorem

- If dp<0, then
- $y_{p+1}=y_p$ and $x_{p+1}=x_p+1$
- $d_{p+1}= (x_p+1)^2+2(x_p+1)+1+y^2_p -y_p+1/4 - r^2$
- $= (x_p+1)^2 +2(x_p+1)+1+(y_p -1/2)^2 - r^2$
- $= (x_p+1)^2 +(y_p -1/2)^2-r^2+2(x_p+1)+1$
- $=d_p+2(x_p+1)+1$
- $=d_p+2x_p+3$

Mid Point Theorem

- If starting point is (0,r), then
- $x_{0=}=0$ and $y_0=r$
- $d_0= (x_0+1)^2+(y_0-1/2)^2 – r^2$
- $= (0+1)^2 +(r-1/2)^2 – r^2$
- $= 1+r^2 – r + ¼ - r^2$
- $= 5/4 - r$

Mid Point Algorithm

- Let us draw a circle with center (xc, yc) and radius r.
- Let (0,r) be starting point.
- X=0, y=r
- p= 1– r
- Putpixels four symmetric axis points.
- While x<=y
- drawpixel(xc,yc,x,y)
- X=x+1
- If p<0, p=p + 2x +3
- Else, p = p + 2(x – y) +5 and y= y- 1

Mid Point Algorithm

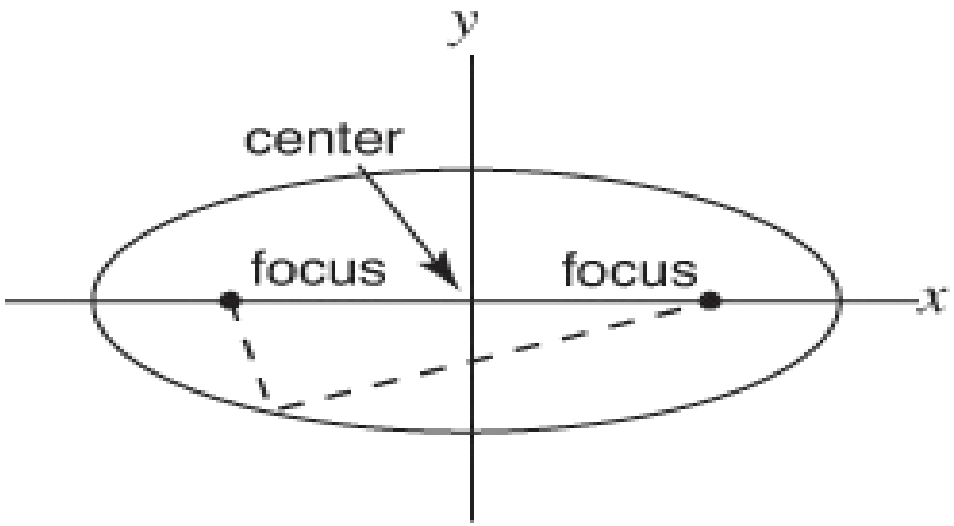
- drawpixel(xc,yc,x,y)
- putpixel(x+xc,y+yc)
- putpixel(x+xc,-y+yc)
- putpixel(-x+xc,y+yc)
- putpixel(-x+xc,-y+yc)
- putpixel(y+xc,x+yc)
- putpixel(y+xc,-x+yc)
- putpixel(-y+xc,x+yc)
- putpixel(-y+xc,-x+yc)

Example

- xc=0, yc=0
- r=10
- x=0, y=r=10
- p=1-10=-9

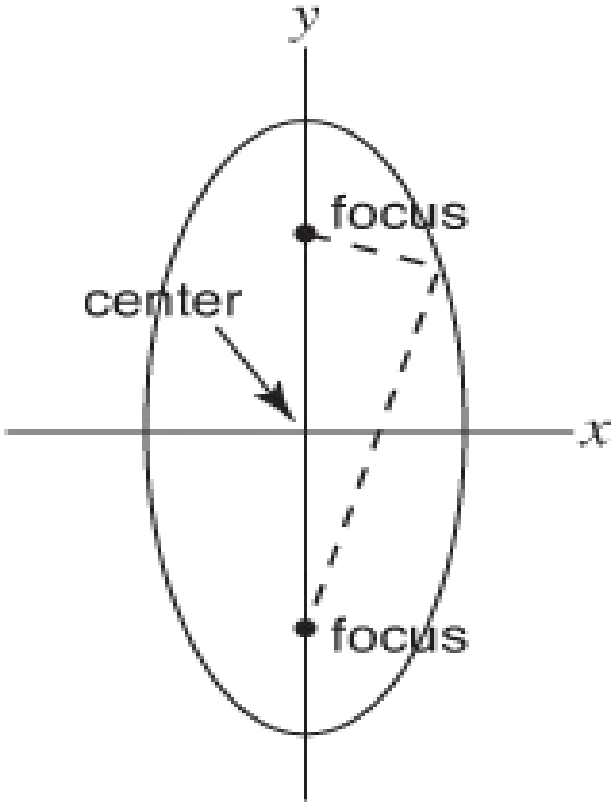
x	y	p	Output
0	10	-9	(0,10)
1	10	-4	(1,10)
2	10	3	(2,10)
3	9	-6	(3,9)
4	9	5	(4,9)
5	8	2	(5,8)
6	7	3	(6,7)
7	6	8	(7,6)
8	5	17	(8,5)
9	4	30	(9,4)
10	3	47	(10,3)

Elipse Algorithm



ellipse with
a horizontal
major axis

(a)

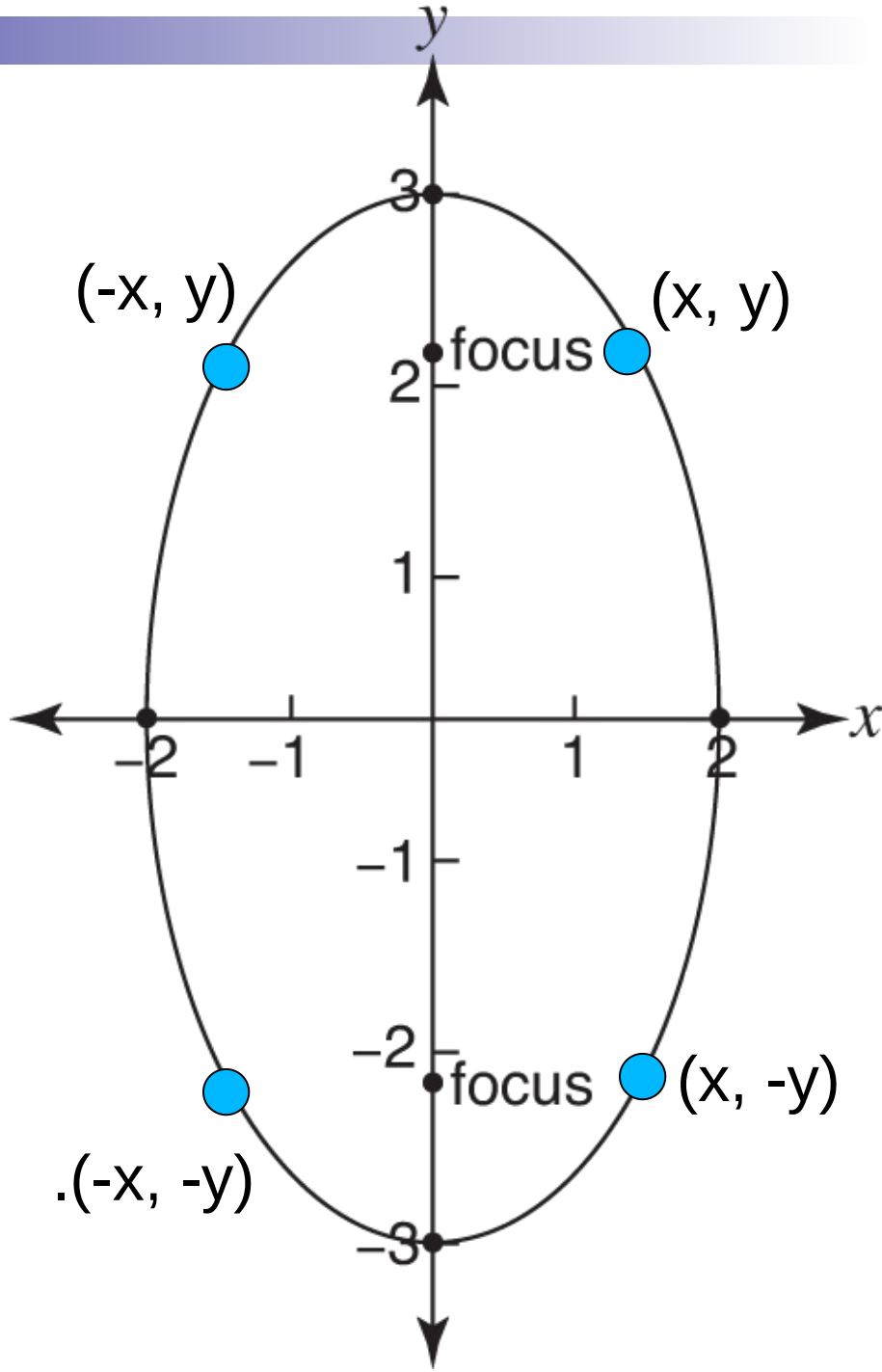


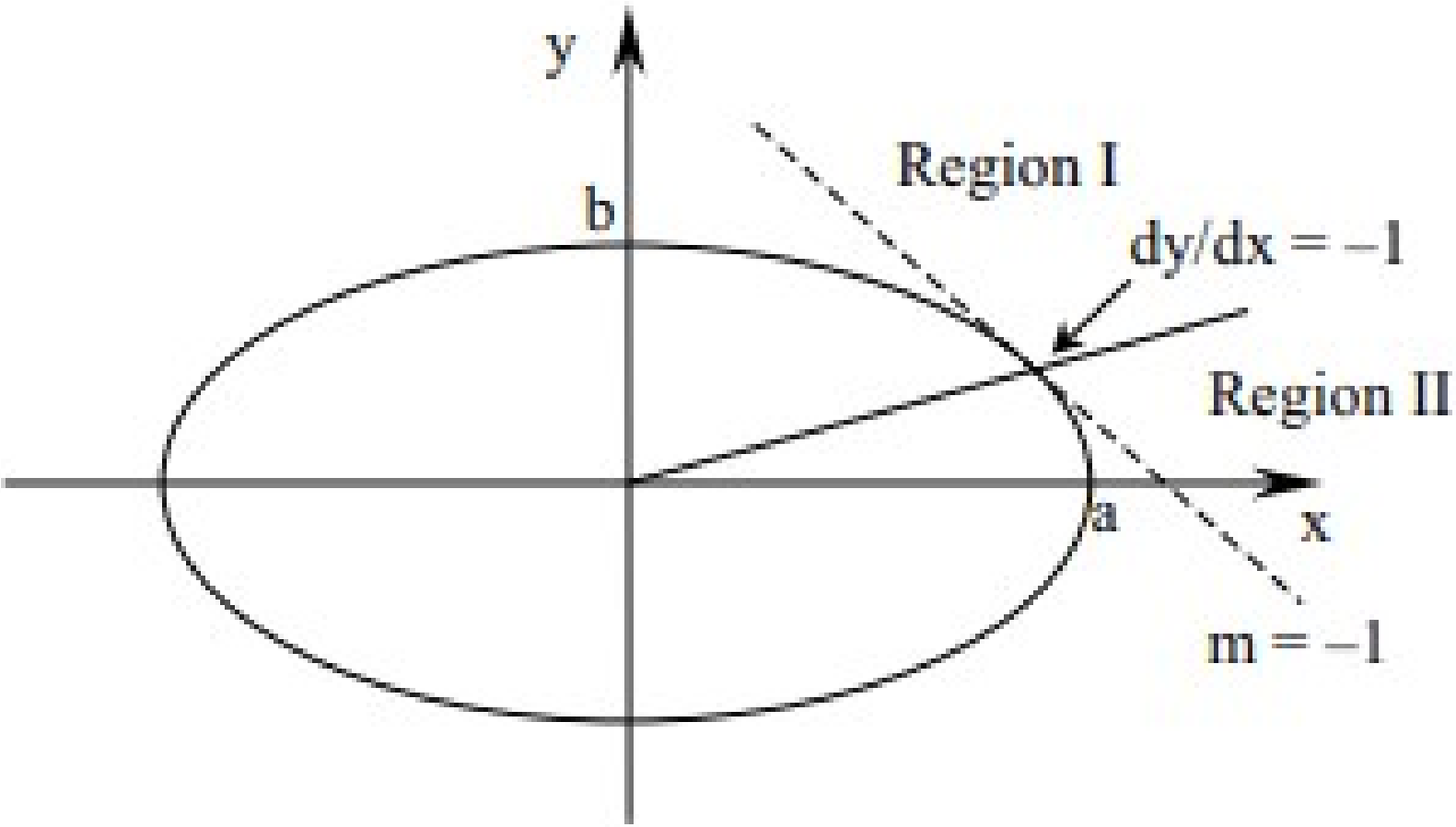
ellipse with
a vertical
major axis

(b)

Elipse Algorithm

- Ellipse is elongated circle.
- Sum of the distances from foci to any point on ellipse is always constant





- Slope at point separating two octants is -1

Center of ellipse $(0, 0)$

Primary axis, a

Secondary axis, b

The equation is

$b^2x^2 + a^2y^2 - a^2b^2 = 0$

Let $f(x, y) = b^2x^2 + a^2y^2 - a^2b^2$

if $f(x, y) > 0, then (x, y)$ lies outside of ellipse.

if $f(x, y) = 0, then (x, y)$ lies on the ellipse.

if $f(x, y) < 0, then (x, y)$ lies inside the ellipse.

The slope at the point that separates two regions is

$\frac{\partial f(x,y)}{\partial y} = \frac{\partial(b^2x^2+a^2y^2-a^2b^2)}{\partial x} = 0$

$b^2 2x + a^2 2y \frac{\partial y}{\partial x} = 0$

$\frac{\partial y}{\partial x} = \frac{-b^2 2x}{a^2 2y}$

if (x, y) is the point that separates two regions, then slope is -1.

$\frac{\partial y}{\partial x} = \frac{-b^2 2x}{a^2 2y} = -1$

$2xb^2 = 2ya^2$

let current point (x_k, y_k) be on the first region.

Now next possible points are $E(x_k + 1, y_k)$ and $SE(x_k + 1, y_k - 1)$.

The midpoint of these two $M(x_k + 1, y_k - 1/2)$.

let $p_k^1 = f(x_k + 1, y_k - 1/2) = b^2(x_k + 1)^2 + a^2(y_k - 1/2)^2 - a^2b^2$

be decision parameter for the first region.

Let $p_{k+1}^1 = b^2(x_{k+1} + 1)^2 + a^2(y_{k+1} - 1/2)^2 - a^2b^2$.

be next decision parameter.

if $p_k^1 > 0, then M$ lies outside, so SE is closer to ellipse.

So, $x_{k+1} = x_k + 1$ and $y_{k+1} = y_k - 1$

$p_{k+1}^1 = b^2(x_{k+1} + 1)^2 + a^2(y_{k+1} - 1/2)^2 - a^2b^2$

$p_{k+1}^1 = b^2(x_k + 1 + 1)^2 + a^2(y_k - 1 - 1/2)^2 - a^2b^2$

$= b^2((x_k + 1)^2 + 2(x_k + 1) + 1) + a^2((y_k - 1/2)^2 - 2(y_k - 1/2) + 1) - a^2b^2$

$= b^2(x_k + 1)^2 + a^2(y_k - 1/2)^2 - a^2b^2 + 2b^2(x_k + 1) + b^2 - 2a^2(y_k - 1/2) + a^2$

$p_{k+1}^1 = p_k^1 + 2b^2(x_k + 1) + b^2 - 2a^2(y_k - 1/2) + a^2$

$p_{k+1}^1 = p_k^1 + 2b^2(x_k + 1) + b^2 - 2a^2y_k + a^2 + a^2$

$p_{k+1}^1 = p_k^1 + 2b^2(x_k + 1) + b^2 - 2a^2(y_k - 1)$

$p_{k+1}^1 = p_k^1 + 2b^2x_{k+1} + b^2 - 2a^2y_{k+1}$

if $p_k^1 < 0, then M$ lies inside, so S is closer to ellipse.

So, $x_{k+1} = x_k + 1$ and $y_{k+1} = y_k$

$$p_{k+1}^1 = b^2(x_{k+1} + 1)^2 + a^2(y_{k+1} - 1/2)^2 - a^2b^2$$
$$p_{k+1}^1 = b^2(x_k + 1 + 1)^2 + a^2(y_k - 1/2)^2 - a^2b^2$$
$$p_{k+1}^1 = b^2((x_k + 1)^2 + 2(x_k + 1) + 1) + a^2(y_k - 1/2)^2 - a^2b^2$$
$$p_{k+1}^1 = b^2(x_k + 1)^2 + 2b^2(x_k + 1) + b^2 + a^2(y_k - 1/2)^2 - a^2b^2$$
$$p_{k+1}^1 = p_k^1 + 2b^2(x_k + 1) + b^2$$
$$p_{k+1}^1 = p_k^1 + 2b^2x_{k+1} + b^2$$

For initial decision parameter, we use initial point $(0, b)$.

$$p_0^1 = b^2(x_0 + 1)^2 + a^2(y_0 - 1/2)^2 - a^2b^2$$
$$p_0^1 = b^2(0 + 1)^2 + a^2(b - 1/2)^2 - a^2b^2$$
$$p_0^1 = b^2 + a^2b^2 - a^2b + a^2/4 - a^2b^2$$
$$p_0^1 = b^2 - a^2b + a^2/4$$

For second region, let current point be (x_k, y_k)
the possible next points are $(x_k, y_k - 1)$ and $(x_k + 1, y_k - 1)$
The mid point is $(x_k + 1/2, y_k - 1)$
The decision parameter is

$$p_k^2 = b^2(x_k + 1/2)^2 + a^2(y_k - 1)^2 - a^2b^2$$

The next decision parameter is

$$p_{k+1}^2 = b^2(x_{k+1} + 1/2)^2 + a^2(y_{k+1} - 1)^2 - a^2b^2$$

for initial decision parameter for second region,
We need to use the last point of the first region on,

$$p_0^2 = b^2(x_0 + 1/2)^2 + a^2(y_0 - 1)^2 - a^2b^2$$

if $p_k^2 > 0$, then $x_{k+1} = x_k$ and $y_{k+1} = y_k - 1$.

$$p_{k+1}^2 = b^2(x_{k+1} + 1/2)^2 + a^2(y_{k+1} - 1)^2 - a^2b^2$$
$$p_{k+1}^2 = b^2(x_k + 1/2)^2 + a^2(y_k - 1 - 1)^2 - a^2b^2$$
$$p_{k+1}^2 = b^2(x_k + 1/2)^2 + a^2((y_k - 1)^2 - 2(y_k - 1) + 1) - a^2b^2$$
$$p_{k+1}^2 = b^2(x_k + 1/2)^2 + a^2(y_k - 1)^2 - a^2b^2 - 2a^2(y_k - 1) + a^2$$
$$p_{k+1}^2 = p_k^2 - 2a^2y_{k+1} + a^2$$

if $p_k^2 < 0$, then $x_{k+1} = x_k + 1$ and $y_{k+1} = y_k - 1$.

$$p_{k+1}^2 = b^2(x_{k+1} + 1/2)^2 + a^2(y_{k+1} - 1)^2 - a^2b^2$$
$$p_{k+1}^2 = b^2(x_k + 1 + 1/2)^2 + a^2(y_k - 1 - 1)^2 - a^2b^2$$
$$p_{k+1}^2 = b^2(x_k + 1/2)^2 + 2b^2(x_k + 1/2) + b^2 + a^2(y_k - 1)^2 - 2a^2(y_k - 1) + a^2 - a^2b^2$$
$$p_{k+1}^2 = p_k^2 + 2b^2(x_k + 1/2) + b^2 - 2a^2(y_k - 1) + a^2$$
$$p_{k+1}^2 = p_k^2 + 2b^2x_k + 2b^2 - 2a^2(y_k - 1) + a^2$$
$$p_{k+1}^2 = p_k^2 + 2b^2(x_k + 1) - 2a^2(y_k - 1) + a^2 + 2b^2$$
$$p_{k+1}^2 = p_k^2 + 2b^2x_{k+1} - 2a^2y_{k+1} + a^2$$

1. start $(0, b)$
2. calculate $p = b^2 - a^2 b$
3. $x = 0, y = b, xc, yc$
4. Repeat:
5. drawellipse(xc, yc, x, y)
6. $x = x + 1$
6. if $p < 0, p = p + 2b^2 x + b^2$
7. else, $y = y - 1, p = p + 2b^2 x + b^2 - 2a^2 y$
8. while $(2b^2 x < 2a^2 y)$
9. $p = b^2 x^2 + a^2 (y - 1)^2 - a^2 b^2$
10. Repeat:
11. drawellipse(xc,yc,x,y)
12. $y = y - 1$
13. if $p > 0, p = p - 2a^2 y + a^2$
14. $else, x = x + 1, p = p + 2b^2 x - 2a^2 y + a^2$
15. while $y >= 0$

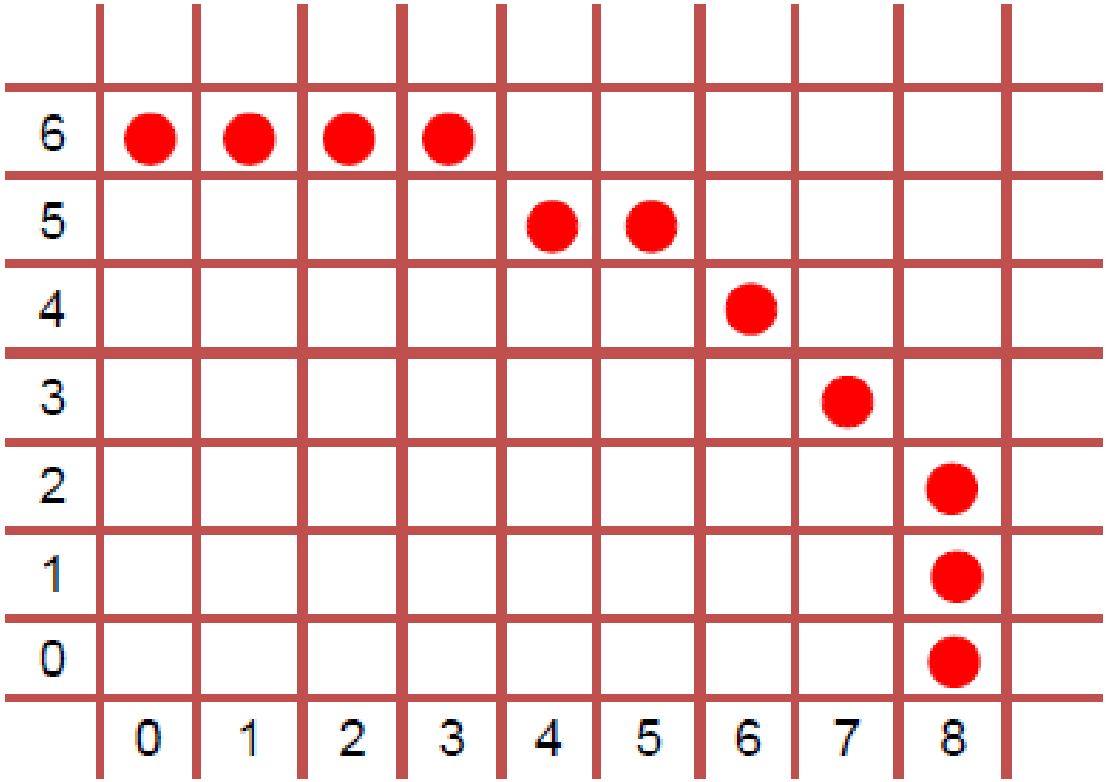
drawellipse(xc, yc, x,y)

1. putpixel(xc+x,yc+y)
2. putpixel(xc+x,yc-y)
3. putpixel(xc-x,yc+y)
4. putpixel(xc-x,yc-y)

Region 1						
a	b	x	y	p	$2b^2x$	$2a^2y$
8	6	0	6	-348	0	768
		1	6	-240	72	768
		2	6	-60	144	768
		3	6	192	216	768
		4	5	-124	288	640
		5	5	272	360	640
		6	4	228	432	512
		7	3	384	504	384

Region 2				
a	b	x	y	p
8	6	7	3	-284
		8	2	-44
		9	1	252
		9	0	316

Plot pixel positions



- Symmetric point calculation:
  - For (0, 6) : (0, 6), (0, -6), (0, -6)
  - For (1,6) : (-1, 6), (-1,-6), (1, -6)
  - For (2,6): (-2,6), (-2, -6), (2,-6)
  - For (3,6): (-3,6), (-3, -6), (3, -6)
  - For(4,5): (-4, 5), (-4,-5) (4,-5)
  - For (5, 5): (-5,5), (-5,-5) (5,-5)
  - For (6,4):(-6,4) (-6,-4) (6,-4)

Area Filling

Instructor: Dinesh Maharjan

2022

Region Filing

- Fills specified region with specified color
- It involves two decision
- The decision of which pixel to fill
- And decision of which color to fill with

Type of Region Filling

- Scan line Polygon filling
- Seed filling (Boundary Fill and Flood Fill)

Seed Fill Algorithm

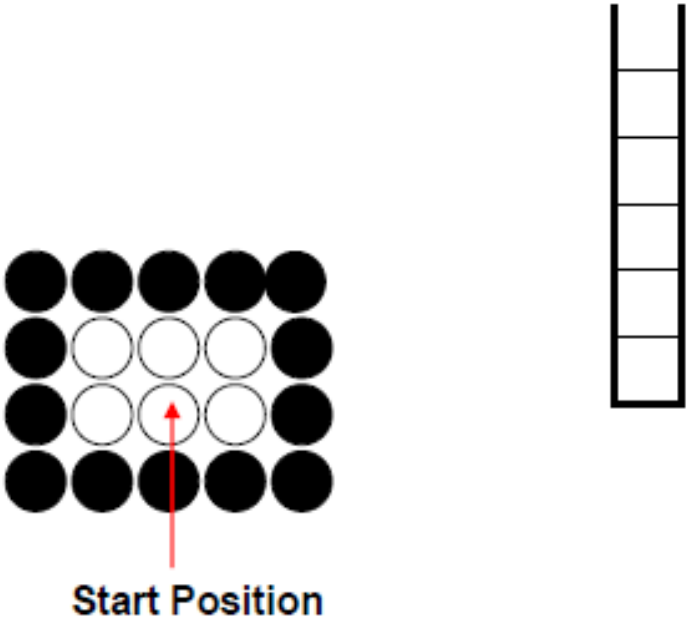
- Simple idea
- Start with one interior pixel and color the region progressively.
- Seed is one of the interior pixel.
- Progressively fills 4 connected or 8 connected pixels of the seed.
- 4 connected are top, down, left, right pixels
- 8 connected are top, top-left, top-right, left, right, bottom, bottom-left, bottom-right

Boundary Fill Algorithm

- Boundary color, fill color, and the seed.
- Push seed to stack.
- repeat.
  - Current pixel=pop stack()
  - Apply specified color to current pixel
  - For each neighboring of seed
    - If neighbor color!= boundary color or neighbor color!=fill color,
      - Push neighboring pixel to stack.
- Until stack is empty

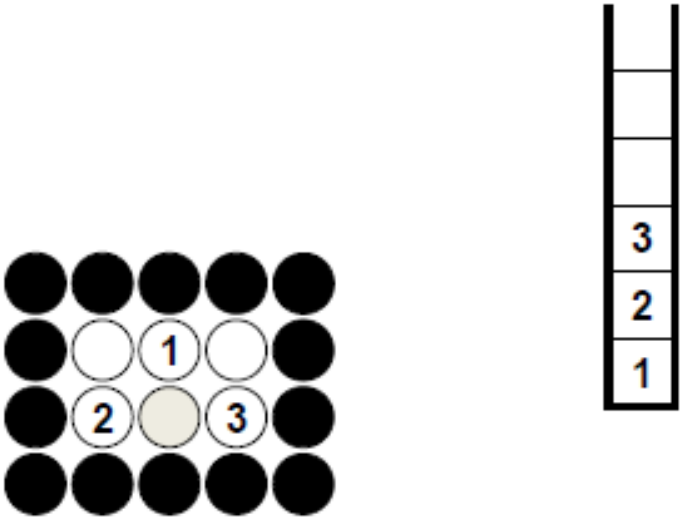
Boundary Fill Algorithm

4-connected (Example)



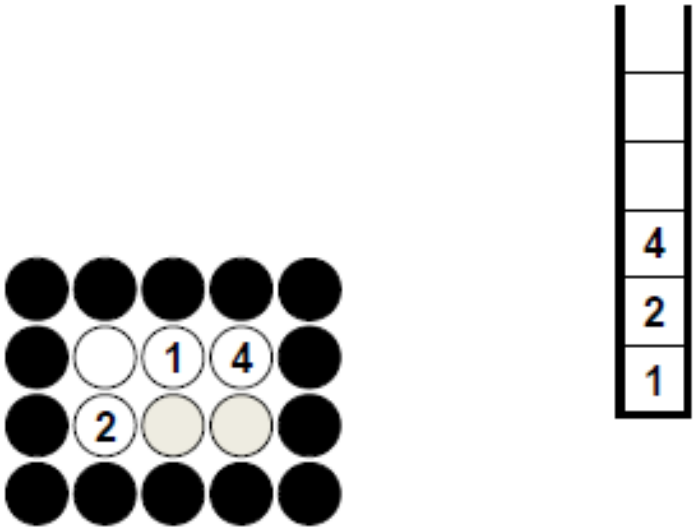
Boundary Fill Algorithm

4-connected (Example)



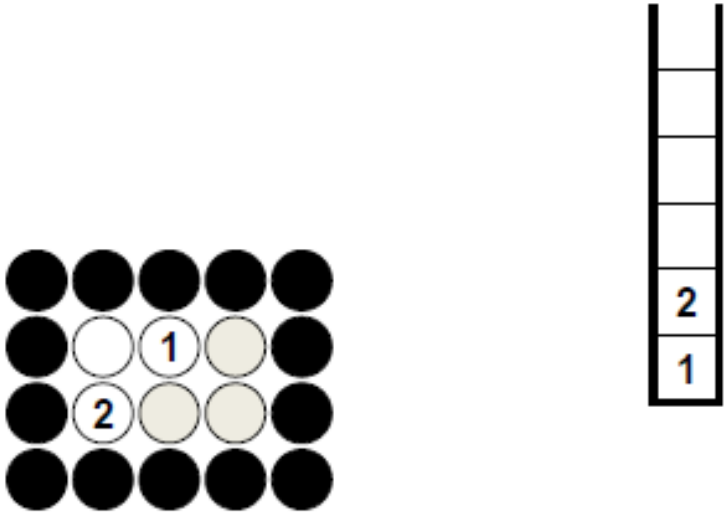
Boundary Fill Algorithm

4-connected (Example)



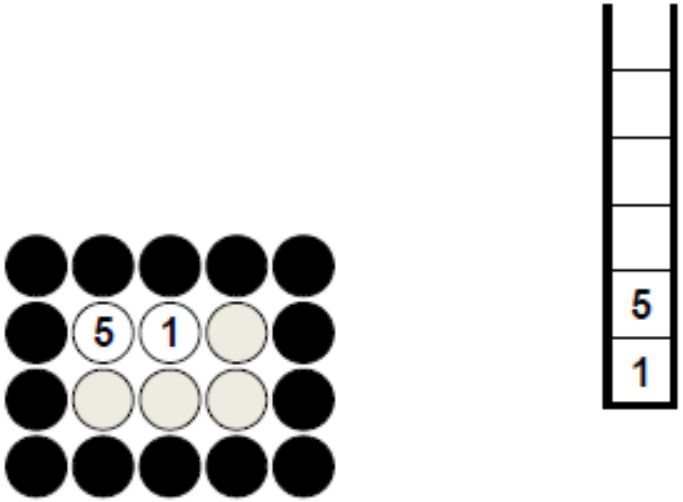
Boundary Fill Algorithm

4-connected (Example)



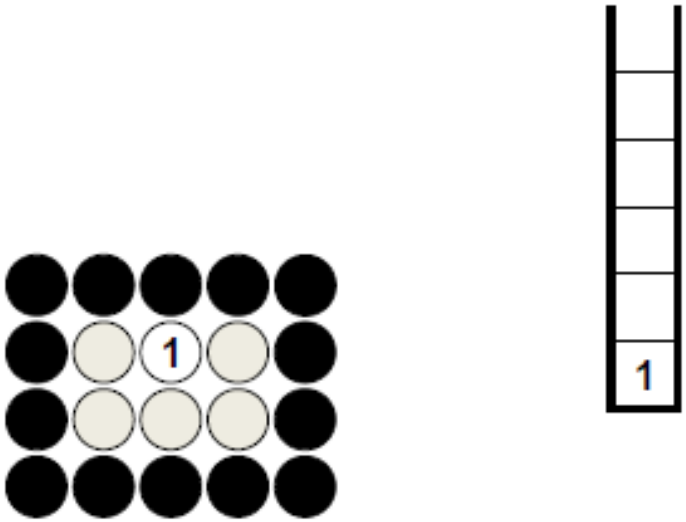
Boundary Fill Algorithm

4-connected (Example)



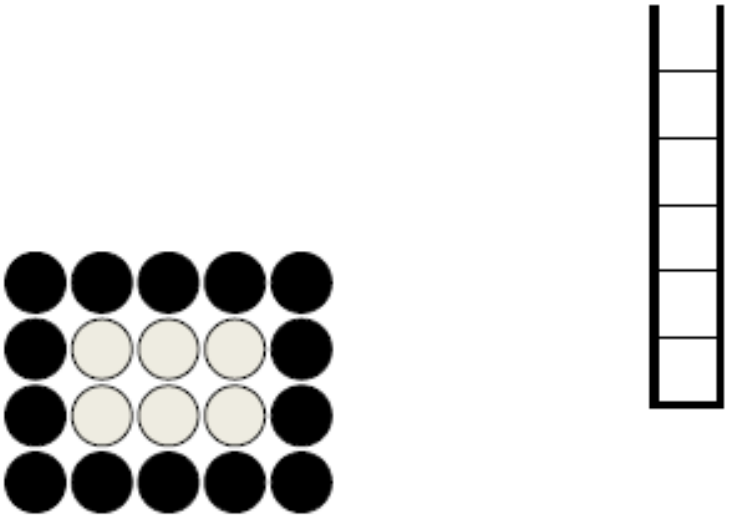
Boundary Fill Algorithm

4-connected (Example)



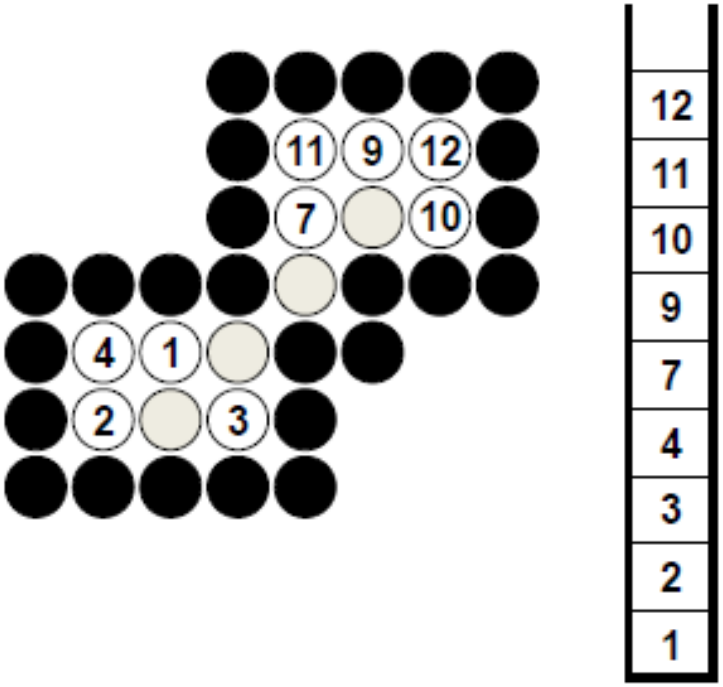
Boundary Fill Algorithm

4-connected (Example)



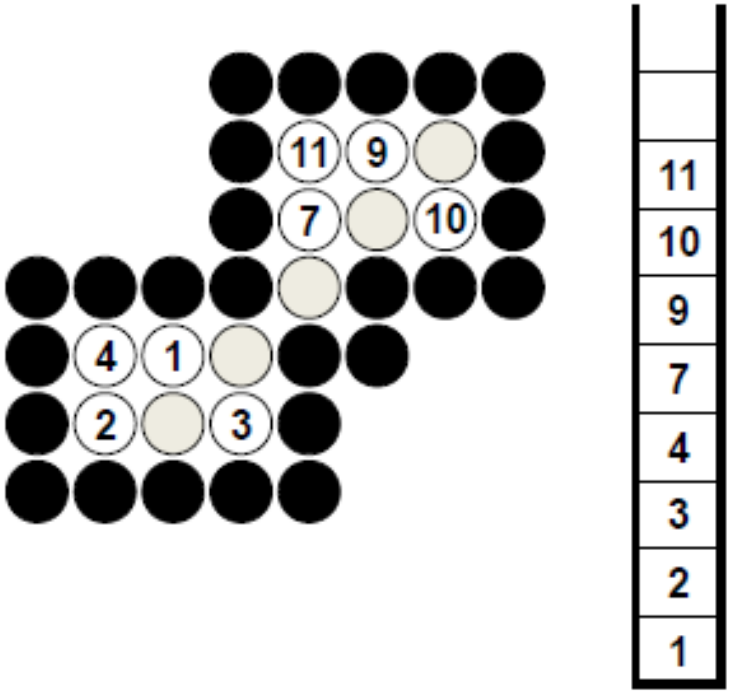
Boundary Fill Algorithm

8-connected (Example)



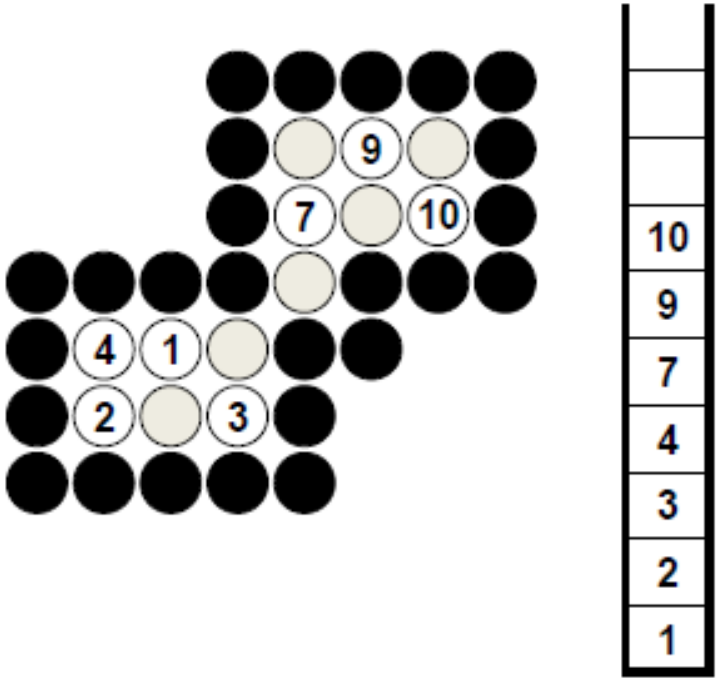
Boundary Fill Algorithm

8-connected (Example)



Boundary Fill Algorithm

8-connected (Example)



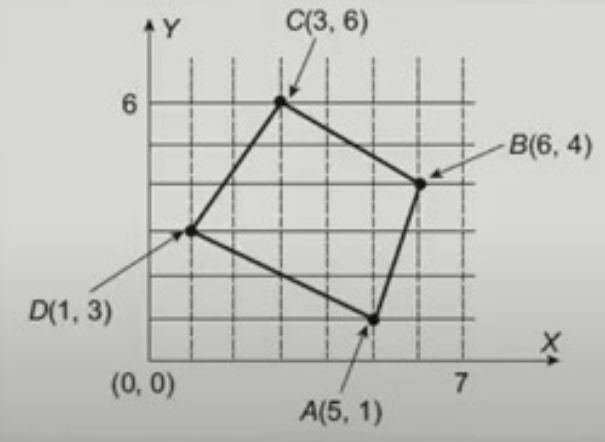
Flood Fill Algorithm

- interior color, fill color, boundary color and the seed.
- Push seed to stack.
- repeat.
  - Current pixel=pop stack()
  - Apply fill color to current pixel
  - For each neighboring pixel of seed
    - If neighbor color == interior pixel
    - Push neighboring pixel to stack.
- Until stack is empty

Scan Line Polygon Algorithm

Illustrative Example

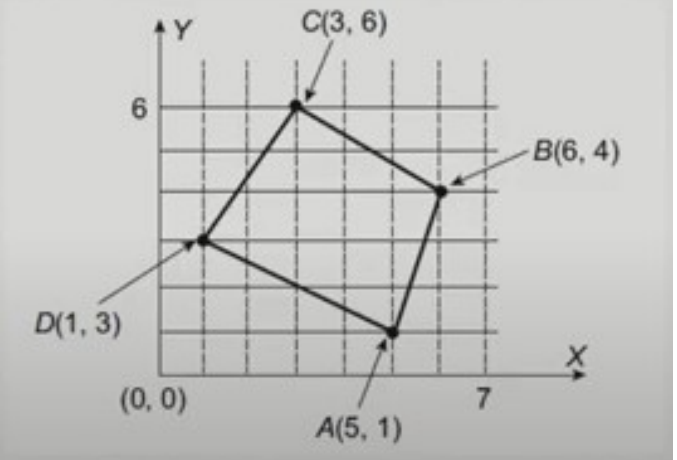
- Polygon specified with vertices *A*, *B*, *C*, and *D* (anti-clockwise vertex naming convention)
- Therefore, edges are *AB*, *BC*, *CD*, and *DA*



Scan Line Polygon Algorithm

Illustrative Example

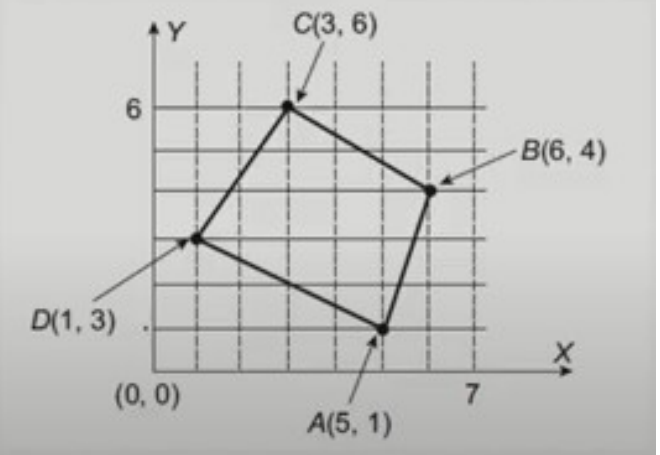
- First, determine max and min scanlines (line 3) from vertex coordinate
  - Max = max{1,4,6,3} = 6
  - Min = min{1,4,6,3} = 1



Scan Line Polygon Algorithm

Illustrative Example

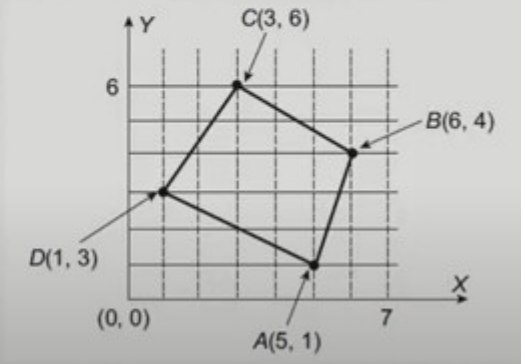
- First iteration of outer loop
  - First determine intersection points of the scan line $y = 1$ with all 4 edges in the inner loop (lines 6–10)



Scan Line Polygon Algorithm

Illustrative Example

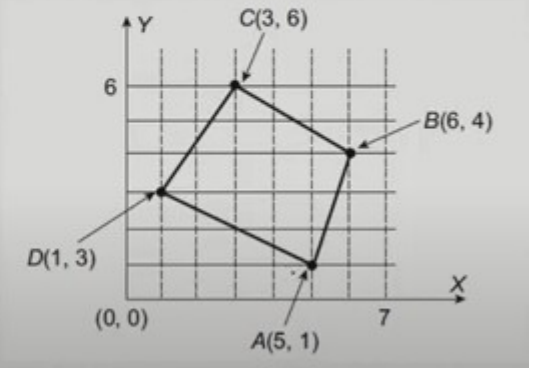
- For *AB*, IF condition satisfied - we determine intersection point as *A* (lines 7–8)
- For *BC* and *CD*, condition not satisfied
- For *DA*, again condition satisfied - we get *A* again



Scan Line Polygon Algorithm

Illustrative Example

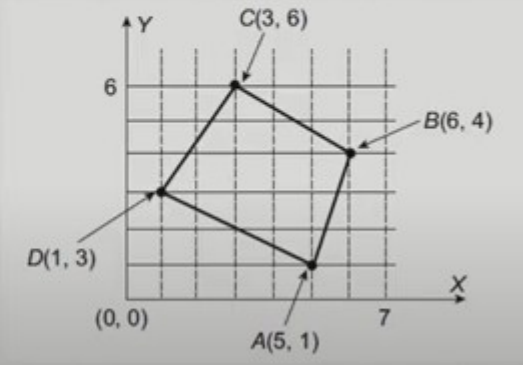
- Thus, 2 intersection points determined are the same vertex *A*
  - This is the only pixel between itself - apply specified color to it (lines 11–12)
- Set scanline = 2 (line 11)
- Since 2 ≠ *max* (= 6), we reenter outer loop



Scan Line Polygon Algorithm

Illustrative Example

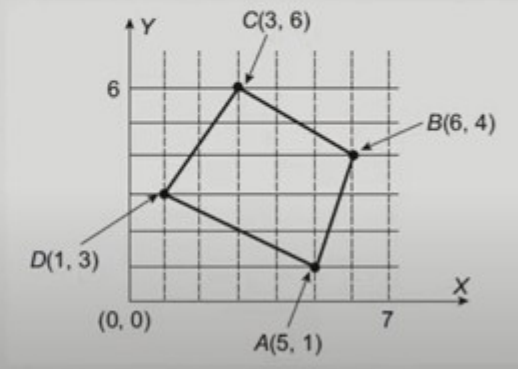
- 2nd iteration of outer loop
  - We check for intersection points between the edges and scanline $y = 2$



Scan Line Polygon Algorithm

Illustrative Example

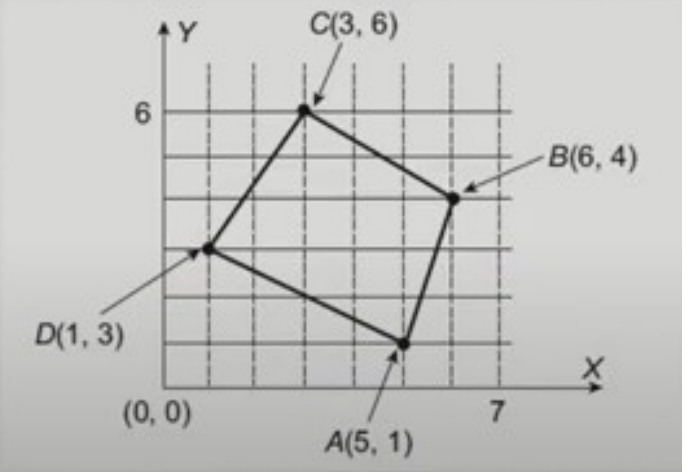
- For *AB*, IF condition satisfied - intersection point $(5\frac{1}{3}, 2)$
- *BC* and *CD* do not satisfy condition – no intersection
- Condition satisfied by *DA* - intersection point is (3,2)



Scan Line Polygon Algorithm

Illustrative Example

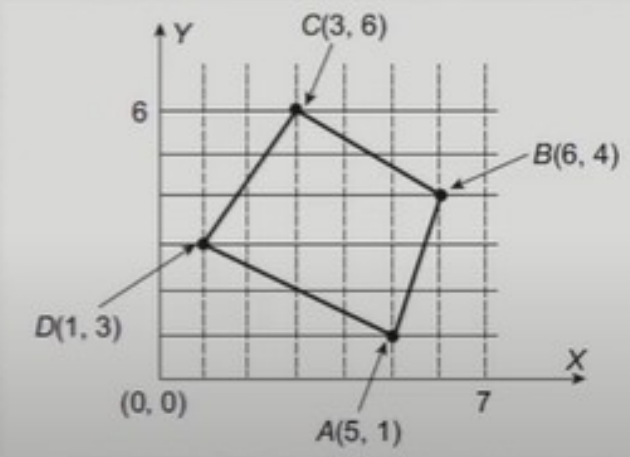
- After sorting (line 11), we get two intersection points (3,2), $(5\frac{1}{3}, 2)$
- Pixels in between - (3,2), (4,2), (5,2)
- Apply specified color to these pixels (line 12) and set scanline = 3
- Since 3 ≠ *max* (= 6), we reenter the outer loop



Scan Line Polygon Algorithm

Illustrative Example

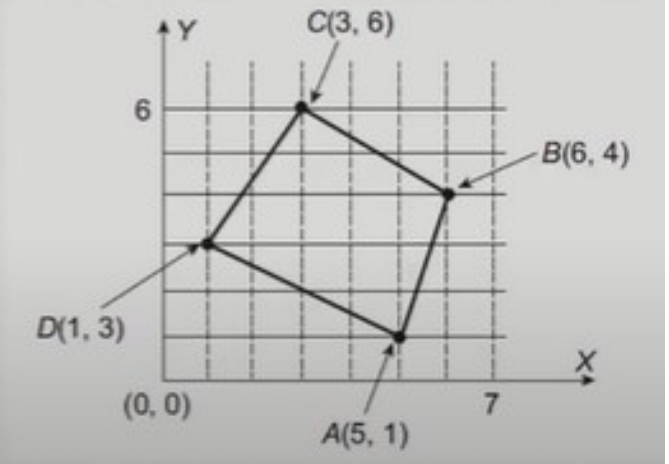
- Algorithm works in similar way for remaining scanlines $y = 3$, $y = 4$, $y = 5$, and $y = 6$



Scan Line Polygon Algorithm

Illustrative Example

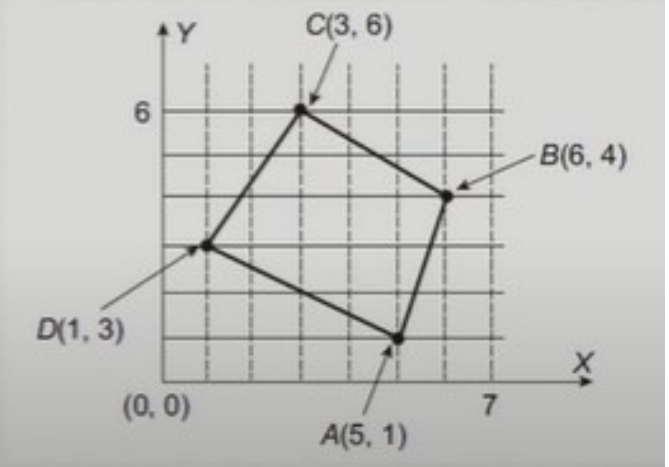
- TWO things in the algorithm require elaboration



Scan Line Polygon Algorithm

Illustrative Example

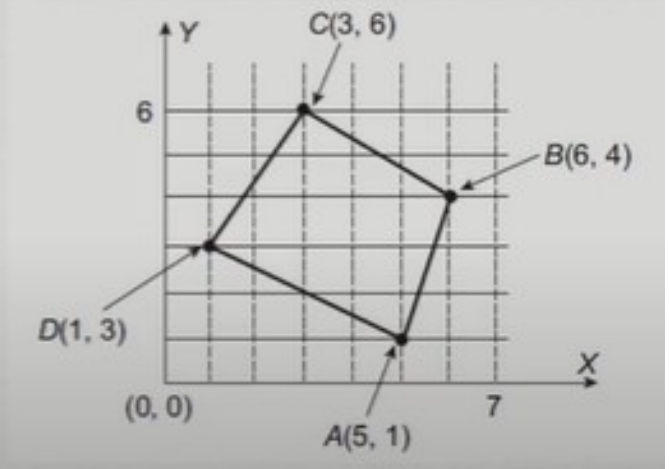
- First, how do we determine the edge–scanline intersection point?
  - Determine line eq from vertices
  - Evaluate with scan line value



Scan Line Polygon Algorithm

Illustrative Example

- Second, how do we determine pixels within two intersection points?
  - Start from left most pixel (x>intersection point)
  - Continue along scan line till x<right intersection



Scan Line Polygon Algorithm

- Is used to fill the polygon region
- Polygon region is the space enclosed by three or more edges.
- So, polygon is defined in terms of vertices and edges.

- Input: Set of vertices, v , specified color, c
- mn=minimum of among y-coordinate from v
- mx=maximum among y-coordinate from v
- s=mn
- Repeat
  - For each edge ((x1,x2), (x2,y2))
    - If (y1<=s<=y2) or (y2<=s<=y1)
      - Find edge and scanline intersection
  - Sort intersection points in increasing order of x-coordinate
  - Apply specified color to the points between intersection points.
- S=s+1
- Until s==mx

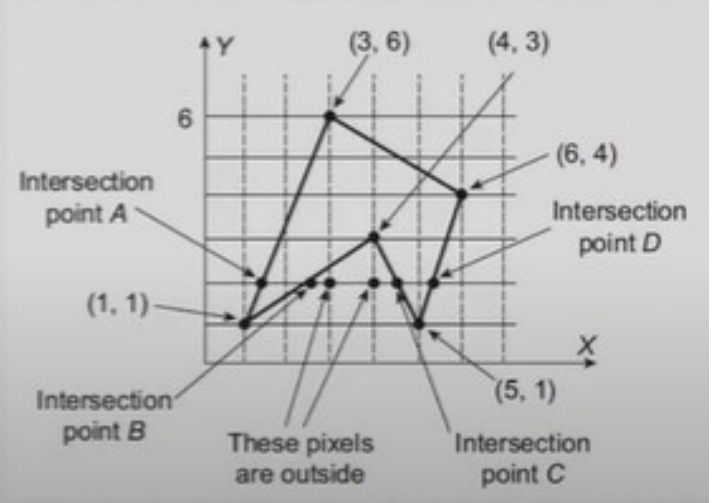
Scan Line Fill for Concave Polygon

- The points between edge and scanline may lie outside if the polygon is concave.
- For such polygon, we need to determine the points that lie inside polygon.
- For testing if a point, p1 lies outside of polygon,
- We need to determine max x and max y from all the vertices of polygon.
- Then choose an arbitrary outside point p2 which is easy as we have max x and max y.

Scan Line Polygon Algorithm

Scan Line Fill for Concave Polygons

- Earlier, we determine pixels between pair of edge–scanline intersection points
  - All these pixels may not be **inside** in case of concave polygon
- We also need to determine **inside** pixels



Scan Line Fill for Concave Polygon

- Join the points p1 and p2.
- If the line cuts the polygon at even number of points then it is outside of polygon.
- Otherwise line is inside of polygon.